

Euclidean Geometry In Mathematical Olympiads

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Problem 1.50 (IMO 2013/4)



IMO

Chapter 1

geometry


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the problem or the solution of the book.

This post has been edited 1 time. Last edited by WalkerV, Nov 9, 2022, 4:17 AM
Reason: Sorry to correct with the mistake of $\$Latex\$$.

rk7697

17 posts

Feb 25, 2023, 11:12 PM

#20

Cool thing I found that might be similar to others: [Click to reveal hidden text](#)

 We can restate the problem by picking a random point W on BC , taking Q as the miquel point from points W, M, N , and showing that connecting H and Q will hit circles (NBW) and (MCW) at X, Y so that $\angle XBW, \angle YCW = 90^\circ$. We can just note that since M, N are the feet of altitudes, $ANHM$ is cyclic, so $(ANQM)$ includes H , and AH is a diameter of $(ANQM)$, so $\angle AQH = 90^\circ$. Notice that $\angle AHN = \angle OHC = \angle NBC = \angle A$ (where O is the foot of the height from a), so $\angle AQN = \angle AHN = \angle A$, and $\angle WQN = \angle WBN = 180 - A$, so $\angle WQA = \angle WQA + \angle HQA = (180 - A) + (A) = 180$, so A, Q, W are collinear and AQ is perpendicular to QH , so $\angle XBW = \angle XQW = 90$ which means WX is a diameter of (BNW) and similarly with WY for (MCW) :]

This post has been edited 5 times. Last edited by rk7697, Feb 26, 2023, 11:57 AM

mudkip42

592 posts

Jun 16, 2025, 8:31 PM

#23

Note that $(ANHM), (BNW), (CMW)$ all intersect at the Miquel Point, P .

Claim: A, P, W are collinear.

Proof: Note that

$$\angle APM = \angle ANM = \angle C = 180^\circ - \angle WPM \implies \angle APW = 180^\circ$$

as desired. ■

Then, since $\angle WPX = \angle WPY = 90^\circ$, X, P, Y are collinear. Then, since $\angle APH = \angle APX = 90^\circ$, H also lies on the same line, so we're done. ■

Attachments:

